Capstone Project Templates

Executive Post Graduate Programme in Generative AI and Agentic AI

Two four-week, six-member team project tracks

Track A — Model-Side (direct model access, GPU required)
Track B — API-Side (hosted ≤8B endpoint, no GPU for teams)

Overview

Parameter	Specification
Team size	6 members
Duration	4 weeks
Assumed effort	~8–10 focused hours per person per week (working professionals)
Curriculum coverage	Modules 1–5: transformers, prompting, RAG, PEFT, multimodal, agents, serving, responsible AI
Tracks	• Track A (GPU, direct model access) • Track B (API only, ≤8B backend)

Both tracks are designed so that the intellectual difficulty sits in the design and evaluation decisions rather than in infrastructure setup. Starter repositories, notebooks and containers are supplied beforehand; the learning is in what teams choose to do with them.

Track A — "Model-Side" Capstone (direct model access, GPU required)

Title

Domain-Adapted Assistant: Fine-Tuning and Serving a Small Open LLM for a Regulated Enterprise Workflow

Scenario

The team picks one narrow enterprise domain — for example insurance claim triage, IT helpdesk resolution, clinical discharge summarisation, bank KYC query handling, or manufacturing SOP question-answering. A general 7–8B instruct model handles the domain poorly: wrong jargon, wrong output schema, inconsistent tone. The team must adapt it and prove the adaptation worked.

What the team must build

1. **Dataset (Week 1).** 800–2,000 instruction–response pairs. Allowed sources: public datasets, synthetic generation from a larger teacher model, or de-identified internal documents. Must include a held-out test set of 150–200 examples the team never trains on.
2. **Baseline (Week 1).** Zero-shot and few-shot prompting of the base model (Llama-3.1-8B-Instruct / Qwen2.5-7B-Instruct / Mistral-7B-Instruct). Record the numbers — this is the bar to beat.
3. **Fine-tuning (Week 2).** QLoRA / LoRA (PEFT) on the base model. At least two runs with different configurations (rank, learning rate, target modules, or epochs) plus one ablation, so the report has a comparison rather than a single lucky run.
4. **Alignment-lite (Week 2–3, optional stretch).** DPO on 200–300 preference pairs, or a rejection-sampling pass — only if Week 2 finished cleanly.
5. **RAG layer (Week 3).** The fine-tuned model wrapped with retrieval over a 200–1,000 document domain corpus. Chunking strategy, embedding model choice and a reranking step must each be a documented decision. Required comparison: base + RAG vs. fine-tuned vs. fine-tuned + RAG.

6. **Evaluation (Week 3).** Task metric (exact match / F1 / ROUGE / schema validity), an LLM-as-judge rubric on the held-out set, and latency plus tokens-per-second measured on the team's own serving stack.
7. **Serving and responsible AI (Week 4).** Merge/export adapters, quantise (GGUF or AWQ/GPTQ), serve via vLLM behind a FastAPI endpoint, containerised. Plus a short model card: intended use, data provenance, failure modes, refusal behaviour, and a red-team log of 20 adversarial prompts.

Role split for six members

Role	Responsibility
Data & synthetic generation	Corpus sourcing, pair generation, de-identification, train/test split
Fine-tuning & tracking	QLoRA runs, hyperparameter sweeps, experiment logging
RAG pipeline	Chunking, embeddings, vector store, reranking
Evaluation & red-teaming	Metric harness, LLM-as-judge rubric, adversarial prompts
Serving, quantisation & infra	Adapter merge, quantisation, vLLM + FastAPI, Docker
Integration & documentation	Repo hygiene, model card, report, demo

Every member runs at least one training job — no one should finish the course without having touched a GPU directly.

Deliverables

- Git repository with reproducible training and serving code
- Experiment tracking dashboard (Weights & Biases or MLflow)
- Model card covering intended use, data provenance, failure modes and refusals
- 12-page technical report
- 10-minute demo of the deployed endpoint

Guardrails that keep it doable in four weeks

- Model size capped at 8B. No pretraining and no full-parameter fine-tuning.
- SSH access to a GPU machine, a working QLoRA starter notebook, and a Dockerfile will be provided.
- Dataset ceiling of 2,000 examples — enough to show real gains, small enough that a run finishes overnight.
- **Week 1 checkpoint is a hard gate:** baseline numbers recorded and dataset frozen. Teams that slip here are instructed to drop the DPO and quantisation stretch goals.

Track B — "API-Side" Capstone (no direct model access, ≤8B model behind an endpoint)

Title

Multi-Agent Operations Copilot: An Agentic System Built Entirely on a Hosted ≤8B Inference API

Scenario

The team is given only an OpenAI-compatible chat-completions endpoint backed by a single shared ≤8B instruct model, plus an embeddings endpoint. They cannot change weights, inspect logits, or swap in a frontier model. Everything must be won through prompting, retrieval, tool design, orchestration and evaluation. This is deliberately the harder engineering discipline — and closer to what most participants will actually do at work.

Suggested domains: procurement and vendor-review assistant, travel-and-expense policy agent, competitive-research analyst, incident postmortem drafter, financial-report question-answering agent, HR policy navigator.

What the team must build

I. **Task decomposition (Week 1).** A written specification of what the copilot does, what it must refuse, and what a correct answer looks like. Plus an evaluation set of 60–100 realistic queries with gold answers, written before any code.

II. **Prompt engineering baseline (Week 1).** Single-agent, single-prompt attempt. Systematic comparison of at least three prompting strategies (structured output with few-shot, chain-of-thought or self-consistency, decomposition prompting). Small models break in specific ways — the team must document how theirs breaks.

III. **RAG over a real corpus (Week 2).** 500–2,000 documents. Required: chunking experiments, hybrid retrieval (BM25 plus dense), a reranker, and citation-grounded answers. Retrieval recall@k must be measured separately from end-to-end answer quality — a key diagnostic when the generator is weak.

IV. **Agentic layer (Week 3).** A multi-agent or tool-using system (LangGraph, CrewAI, AutoGen, or hand-rolled) with at least three tools, an explicit planner→executor→critic loop, persistent state across turns, and error handling for tool failures, malformed JSON, infinite loops and token budget caps.

V. **Evaluation and observability (Week 3–4).** End-to-end accuracy against the Week 1 evaluation set, task completion rate, average tool calls per task, cost and latency per query, and a failure taxonomy separating retrieval miss, reasoning error, tool-call error and hallucination. Tracing via LangSmith, Langfuse or OpenTelemetry.

VI. **Ship it (Week 4).** Streamlit or React front-end, containerised, with rate limiting and caching. Responsible-AI section covering prompt-injection tests against the document corpus, PII handling, refusal behaviour, and a stated human-in-the-loop boundary for consequential actions.

Small models fumble tool calls, so constrained decoding, JSON schema validation and retry logic are the core engineering content of this track.

Role split for six members

Role	Responsibility
Domain spec & eval design	Scope definition, gold-answer evaluation set, refusal policy
Prompt engineering	Strategy comparison, structured output, failure characterisation
RAG / retrieval	Chunking, hybrid search, reranking, citation grounding
Agent orchestration & tools	Planner/executor/critic loop, tool interfaces, state and memory
Observability & safety	Tracing, failure taxonomy, prompt-injection and PII testing
Front-end & deployment	UI, containerisation, rate limiting, caching, demo

Deliverables

- Git repository with the full agent stack
- Trace dashboard showing real query executions
- Evaluation report including the failure taxonomy
- 12-page technical report
- 10-minute live demo

Guardrails that keep it doable in four weeks

- The endpoint, an API key with a per-team token quota, and a starter agent skeleton will be provided.
- Corpus capped at 2,000 documents; agent capped at 10 steps per query.
- **Week 1 gate:** evaluation set frozen and baseline numbers recorded before any agent code is written. This is the single most common failure mode — teams that build first and evaluate later have nothing to report.
- The weakness of the 8B model is an explicit learning objective, not a complaint. “What did we do to compensate for a small model?” is a graded section.

Side-by-Side Comparison

	Track A (GPU)	Track B (API)
Modules emphasised	1, 3, 5 (and 2)	2, 4, 5
Core skill	Model adaptation and serving	Orchestration and evaluation
Hardest week	Week 2 (training instability)	Week 3 (agent reliability)

Grading

Component	Weight
Technical depth and rigour	30%
Punctuality and consistent progress	25%
Working deployed system	20%
Report and demo	25%